Orchestration design

Multi-Agent AI for Smart Contract Audit
and Cybersecurity Risk Assessment

Workspace flow. Run Bedrock Review, inspect Findings, then open Second Review and select Run Second Review. Final Report follows an accepted second-round result or a stated reason that no review is eligible. No second round starts automatically.

GPT-6.1 Sol, Claude Opus 5.5 and Grok 4.7 provide judgments through Amazon Bedrock. A deterministic supervisor schedules their work, validates evidence and consolidates outputs. It adds no fourth model.

Two iterations. Traceable assignments.

Stage	Input and responsibility	Retained output
1. Independent review	Same numbered source and prompt variant. No static findings or peer answers.	Original findings, execution outcomes and independent agreement.
2. Targeted plan	Select up to 12 queued claims. Assign peer findings a reviewer omitted; also reassess severity disputes.	Exact IDs per model, unassigned reviewers and deferred claims.
3. Second Review	Explicit user action sends source and assignments. Empty assignments make no call; failed initial models remain unavailable.	Support, rejection or uncertainty, with rationale and exact evidence.
4. Final Report	Read risk explanations and actions first; expand model judgments and evidence. Preserve both original rounds.	First discoverers, later decisions, coverage and separate severity alignment.

Typed decisions, not automatic truth

The JEV reading motivates constrained decisions. A stable 12-slot grammar holds assigned judgments and null unused slots. The server validates every slot before binding original claim IDs. All models cite source spans; exact text is retrieved after validation.

Peer omissions are the primary assignment rule; severity reassessment is an extension. Failed reassessment retains a valid original opinion and severity as first-round evidence, with pending work explicit. It adds no new vote.

A bounded decision process

One final view, with original evidence and later judgments kept distinct.

Effective positions across three models	Final disposition
All three support, using labeled first- or second-round evidence	Supported; severity differences and pending reassessment still require review
All three explicitly do not support	Not supported
Missing, mixed or uncertain judgments, or deferred work	Unresolved; retain for human review

Final Report retains the last valid opinion. An accepted second-round decision supersedes it; a failed reassessment preserves original evidence and remains pending. Partial execution is shown explicitly. Cancelled requests without a report do not unlock Final Report.

Execution contract

Boundary	Enforced limit
Model / pass	At most 3 attempts within one absolute 270-second deadline
Server request	285 seconds; 295-second browser wait; 300-second function; separate requests per pass
Second-round work	One round; up to 12 claims; only nonempty assignments to successful first-pass models
Provider calls	Up to 9 per pass; 18 per contract; 90 for five paired cases

Each round allows three attempts per model, shared by temporary-error or remote-model-timeout recovery and local validation correction. Safe constraints request a complete new answer; rejected text is never echoed or accepted. Filtering, access, credential echo, provider JSON parsing and local cancellation stay terminal. Successful models are not rerun and no deadline resets.

Second Review uses the exact baseline; hashes are not signatures. A stable native schema avoids assignment-count changes; latency is not guaranteed. Grok uses medium effort. Keys stay in tab/request memory. Accepted AWS work may be billed; guards are not global quotas.

Evaluate the same evidence

Keep the five original contracts fixed and pair each second round with its exact first-round baseline.

Question	Required evidence
Does it execute?	Completion, attempts, validation failures, latency, available usage and unstarted work.
Did opinions change?	Same claim and baseline; original discovery versus later support, rejection or uncertainty.
Are findings correct?	Human labels tied to source and baseline: confirmed, refuted, uncertain or unreviewed.
What was missed?	A complete reviewed issue inventory, including issues absent from every model report.
Did review regress?	New false positives and lost true findings alongside any gains; retain coverage gaps.

Practical evaluation

1. Open Review settings for five-case validation. Select its own follow-up setting and run sequentially. Export all outcomes, including filtered, failed, skipped and deferred work.

2. Review distinct root causes and locations independently of model agreement. Import source-bound adjudication data; leave uncertain or unreviewed claims explicit.

3. Compare eligible paired counts. The final view's retained opinions do not count as successful second-round judgments. Precision requires reviewed predictions; recall also requires a complete issue inventory.

The final report supports risk triage and traceability in existing security workflows. Higher agreement alone is not improved accuracy. These five teaching fixtures are not a held-out benchmark; simulated tests establish implementation behavior, not live model quality.

JEV reading I Documentation I Platform overview